CORE PURE / FURTHER CALCULUS

Improper Integrals and Mean Values

Mark scheme 16 questions · 116 marks

Questions

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	Total	**116**	

Reading the mark scheme

These rules apply throughout, unless a note says otherwise.

Marks

Every mark is worth one, so **M1 A1** on a row is two marks. Only marks coded ft follow through.

Code	Meaning
M1	Method: a correct method, applied to this question. A slip doesn't cost it if the method is clear. A formula must be used, not just quoted.
dM1	Dependent method: as M1, but only if the method mark before it was earned, unless its note names another.
A1	Accuracy: a correct answer or step. It needs the method mark it follows.
A1ft	As A1, following through from an earlier answer that the note names.
A1 AG	The last mark of a part whose answer is printed in the question: complete working, with no errors.
B1	A correct value, fact or statement. It needs no method mark.
B1ft	As B1, following through from an earlier answer that the note names.
E1	A written explanation, interpretation or conclusion. Concluding a proof by induction needs every earlier mark.

Abbreviations

ft and AG qualify a code in the Marks column. The others appear in the notes.

ft	follow through	**AG**	answer given in the question
cao	correct answer only, with no follow-through	**cso**	correct solution only: no errors in the part
oe	or an equivalent form	**awrt**	any answer that rounds to this value
soi	seen, or implied by later working	**isw**	ignore working after a correct answer
SC	special case: credit outside the main method	**their**	the student's own earlier value
condone	a slip that doesn't cost the mark	[ ]	working the mark doesn't need
s.f.	significant figures	**d.p.**	decimal places

Reading the scheme

- Each row is a line of working, with the marks it earns beside it. Where a row carries two marks, each note starts with its code.
- **[*n*]** ends each part with its total.
- A shaded *Alternative* is a second method for the same total. Mark an answer by the method it follows.
- Any other valid method earns the equivalent marks, unless the question names a method or says *hence*.
- A sketch earns one B1 for each feature named below its figure.

Accuracy and answers

- Unless a question says otherwise, give answers to 3 significant figures, and angles in degrees to 1 decimal place. Angles in radians take 3 significant figures.
- A final answer with more figures is accepted if it rounds to the scheme's value; one with fewer is not. An answer spoiled by rounding too early loses its accuracy mark.
- Where a question asks for an exact answer, a decimal loses the final accuracy mark. Otherwise either is accepted.
- Wrong or missing units cost nothing unless the question asks for them.
- An omitted $\mathrm{d}\theta$, $\mathrm{d}x$ or $\mathrm{d}u$ in an integral is not penalised where the variable of integration is clear from the integrand and the limits. It is penalised where it is genuinely ambiguous: a substitution part-way through, two variables in play, or limits in the wrong variable.
- A correct answer with no working earns full marks, unless the question needs the working: show that, prove, hence, a method it names, or an exact answer. A wrong answer with no working earns nothing. Calculator answers are treated the same way.
- A correct answer, once reached, isn't spoiled by later working (isw).

Value of g

- Take $g = 9.8\,\mathrm{m\,s^{-2}}$ unless a question states a value, which is then the only one accepted. Answers that use $g = 9.8$ may be given to 2 or 3 significant figures.
- Where a question doesn't state g, an answer worked with $g = 10$ throughout earns the same marks, keeping any value printed in the question. The worked solutions list the final answers with $g = 10$.

Question	Scheme	Marks	Notes
1	$\int \frac{x}{x^2+1}\,\mathrm{d}x = \frac{1}{2}\ln(x^2+1)$	**M1 A1**	**M1** Integrates to the form $k\ln(x^2+1)$, or $k\ln u$ after substituting $u = x^2+1$. **A1** oe, such as $\frac{1}{2}\ln u$. Condone modulus signs.
	$\frac{1}{3-1}\left(\text{“}\frac{1}{2}\text{”}\ln 10 - \text{“}\frac{1}{2}\text{”}\ln 2\right)$	**dM1**	Substitutes limits 1 and 3 (or 2 and 10 in u) into their $k\ln(x^2+1)$, subtracts, and divides by 2.
	$\frac{1}{4}\ln 5$	**A1**	Must be in the form $p\ln q$; accept $\frac{1}{8}\ln 25$. A0 for $\frac{1}{4}(\ln 10 - \ln 2)$ or a decimal.
	Total for Question 1	**4**	

Question	Scheme	Marks	Notes
2	$\int x(x^2+5)^{-\frac{3}{2}}\,\mathrm{d}x = k(x^2+5)^{-\frac{1}{2}}$	**M1**	Integrates to the form $k(x^2+5)^{-\frac{1}{2}}$, $k \neq 0$, by reverse chain rule or substitution $u = x^2+5$.
	$= -(x^2+5)^{-\frac{1}{2}}\ (+c)$	**A1**	oe, such as $-\frac{1}{\sqrt{u}}$ with $u = x^2+5$.
	$\int_2^t \ldots \mathrm{d}x = \left[-\frac{1}{\sqrt{x^2+5}}\right]_2^t$ $= -\frac{1}{\sqrt{t^2+5}} + \frac{1}{3}$	**M1**	Replaces ∞ by t (any letter) and substitutes t and 2, or corresponding limits in u, into their $k(x^2+5)^{-\frac{1}{2}}$, subtracting.
	As $t \to \infty$, $\frac{1}{\sqrt{t^2+5}} \to 0$	**dM1**	States that their term in t tends to 0 as $t \to \infty$, in limit notation or words. Not for substituting ∞ directly.
	$\int_2^\infty \ldots \mathrm{d}x = \frac{1}{3} - 0 = \frac{1}{3}$	**A1 AG**	cso: correct antiderivative and limits leading without error to $\frac{1}{3}$, with the limiting process shown.
	Total for Question 2	**5**	

Question	Scheme	Marks	Notes
3	$\dfrac{4}{x(x+1)(x+2)} = \dfrac{A}{x} + \dfrac{B}{x+1} + \dfrac{C}{x+2}$	**M1**	Uses the correct form with three constants. Not for a form missing a term.
	$4 = A(x+1)(x+2) + Bx(x+2) + Cx(x+1)$ $\Rightarrow A = \ldots,\ B = \ldots,\ C = \ldots$	**dM1**	Finds all three constants, by substituting values or equating coefficients. Condone arithmetic slips.
	$\dfrac{2}{x} - \dfrac{4}{x+1} + \dfrac{2}{x+2}$	**A1**	
	$2\ln x - 4\ln(x+1) + 2\ln(x+2)$	**M1 A1**	**M1** Integrates at least two of their partial fractions of the form $\frac{a}{x+b}$ to $a\ln(x+b)$. Modulus signs not needed. **A1** All three correct; need not be combined.
	$\left[2\ln\dfrac{x(x+2)}{(x+1)^2}\right]_1^t$ $= 2\ln\dfrac{t(t+2)}{(t+1)^2} - 2\ln\dfrac{3}{4}$	**M1**	Replaces ∞ by t or uses a limit, combines their log terms into one logarithm of a function of t, and substitutes both limits. Condone sign slips.
	As $t \to \infty$, $\dfrac{t^2+2t}{t^2+2t+1} = \dfrac{1+\frac{2}{t}}{1+\frac{2}{t}+\frac{1}{t^2}} \to 1$ $\Rightarrow 2\ln(\ldots) \to 0$, so the integral converges	**dM1**	Shows the argument of the combined logarithm tends to 1 (e.g. dividing by t^2), so its log tends to 0, and concludes convergence. Not for 0 written without a limit argument.
	$\ln\dfrac{16}{9}$	**A1**	cso. Must be in the form $\ln k$, or $k = \frac{16}{9}$ stated; $2\ln\frac{4}{3}$ alone does not score.
	Total for Question 3	**8**	

Question	Scheme	Marks	Notes
4	$y = \dfrac{1+\sin x}{1-\sin^2 x} = \dfrac{1+\sin x}{\cos^2 x}$	**M1**	Multiplies numerator and denominator by $1+\sin x$ and uses $1-\sin^2 x = \cos^2 x$ to reach a denominator of $\cos^2 x$.
	$y = \sec^2 x + \sec x \tan x$	**A1**	
	$\displaystyle\int \left(\sec^2 x + \sec x \tan x\right) \mathrm{d}x = \tan x + \sec x$	**M1 A1**	**M1** Integrates their expression of the form $p\sec^2 x + q\sec x \tan x$ to the form $a\tan x + b\sec x$. **A1** A1: Correct $\tan x + \sec x$.
	$\dfrac{6}{\pi}\left[\left(\frac{1}{\sqrt{3}} + \frac{2}{\sqrt{3}}\right) - (0+1)\right]$	**dM1**	Substitutes both limits into their integral, subtracts, and divides by $\frac{\pi}{6}$. Need not be simplified.
	$\dfrac{6\left(\sqrt{3}-1\right)}{\pi}$	**A1**	cso. Any equivalent exact form.
	Total for Question 4	**6**	

Question	Scheme	Marks	Notes
5	$\frac{1}{2}\int_0^2 40\tanh^2\left(\frac{1}{2}t\right)\mathrm{d}t$ $= 20\int_0^2\left(1-\operatorname{sech}^2\left(\frac{1}{2}t\right)\right)\mathrm{d}t$	**M1**	Uses a hyperbolic identity, such as $\tanh^2 x = 1-\operatorname{sech}^2 x$, to replace $\tanh^2\left(\frac{1}{2}t\right)$ in the integral of p. Condone a sign slip in the identity.
	$20\left[t-2\tanh\left(\frac{1}{2}t\right)\right]_0^2$	**M1 A1**	**M1** Integrates to the form $\alpha t+\beta\tanh\left(\frac{1}{2}t\right)$. **A1** Correct, with factor 20, or 40 if not yet divided by 2.
	$20\left(2-2\tanh 1\right)-20(0)\ [=40(1-\tanh 1)]$	**dM1**	Substitutes both limits into their integral and divides by 2. Lower limit giving 0 need not be shown.
	$\tanh 1=\frac{e-e^{-1}}{e+e^{-1}}\ \left[=\frac{e^2-1}{e^2+1}\right]$	**M1**	Writes $\tanh 1$, or $\sinh 1$ and $\cosh 1$, in exponential form in their expression. Need not be simplified.
	$\frac{80}{e^2+1}$	**A1**	cso. Only this form scores, not e.g. $\frac{80e^{-2}}{1+e^{-2}}$ or $40(1-\tanh 1)$.
	Total for Question 5	**6**	

Question	Scheme	Marks	Notes
6(a)	Upper limit is infinite, so the interval of integration is unbounded	**E1**	States that the upper limit is infinite, or the interval unbounded. Not for a reason about the integrand alone.
		[1]	
6(b)	$u = \ln x,\ \frac{\mathrm{d}v}{\mathrm{d}x} = x^{-3} \Rightarrow v = -\frac{1}{2x^2}$ $\int \frac{\ln x}{x^3}\,\mathrm{d}x = -\frac{\ln x}{2x^2} + \int \frac{1}{2x^3}\,\mathrm{d}x$	**M1 A1**	**M1** Integrates by parts with $u = \ln x$, reaching the form $px^{-2}\ln x \pm \int qx^{-3}\,\mathrm{d}x$. Condone sign and coefficient slips. **A1** Correct first stage; need not be simplified.
	$= -\frac{\ln x}{2x^2} - \frac{1}{4x^2} + c$	**A1**	oe. Needs $+c$.
		[3]	
6(c)	$\int_e^t \frac{\ln x}{x^3}\,\mathrm{d}x = \left[\text{“}-\frac{\ln x}{2x^2} - \frac{1}{4x^2}\text{”}\right]_e^t$	**M1**	Replaces ∞ by t and substitutes t and e into their (b) of the form $px^{-2}\ln x + qx^{-2}$. Not for substituting ∞ directly.
	$= \frac{3}{4e^2} - \frac{\ln t}{2t^2} - \frac{1}{4t^2}$	**A1**	oe; $\ln e$ need not be evaluated.
	As $t \to \infty$, $\frac{\ln t}{t^2} \to 0$ and $\frac{1}{t^2} \to 0$	**dM1**	Lets $t \to \infty$, stating explicitly that $\frac{\ln t}{t^2} \to 0$ (oe) and $\frac{1}{t^2} \to 0$. Not for replacing the terms by 0 without a statement.
	Converges: $\int_e^\infty \frac{\ln x}{x^3}\,\mathrm{d}x = \frac{3}{4e^2}$	**A1**	cso. Exact value oe, with a statement that the integral converges or the limit is finite.
		[4]	
	Total for Question 6	**8**	

Question	Scheme	Marks	Notes
7(a)	$9(x^2-2x)+25=9\left[(x-1)^2-1\right]+25$	**M1**	Takes out 9 from the x terms and completes the square to $9(x\pm1)^2+q$, or compares coefficients of $a(x+p)^2+q$. Condone a slip in q.
	$=9(x-1)^2+16$	**A1**	Or $a=9$, $p=-1$, $q=16$ stated.
		[2]	
7(b)	$\int \frac{\mathrm{d}x}{\sqrt{9(x-1)^2+16}}$ $=\frac{1}{3}\int \frac{\mathrm{d}x}{\sqrt{(x-1)^2+\left(\frac{4}{3}\right)^2}}$	**M1**	Uses their (a), of the form $9(x+p)^2+q$ with $q>0$, with the arsinh result to reach $\lambda\,\mathrm{arsinh}(\mu(x+p))$ or a logarithmic equivalent. Condone slips in λ, μ.
	$=\frac{1}{3}\,\mathrm{arsinh}\left(\frac{3(x-1)}{4}\right)+c$	**A1**	oe, including a logarithmic form. Needs $+c$.
		[2]	
7(c)	$\frac{1}{2}\times\frac{1}{3}\left[\mathrm{arsinh}\left(\frac{3(x-1)}{4}\right)\right]_0^2$ $=\frac{1}{6}\left(\mathrm{arsinh}\,\frac{3}{4}-\mathrm{arsinh}\left(-\frac{3}{4}\right)\right)$	**M1**	Substitutes 0 and 2 into their (b), of the form $\lambda\,\mathrm{arsinh}(\mu(x+p))$ or a logarithmic equivalent, subtracts and divides by 2. Condone sign slips.
	$\mathrm{arsinh}\,\frac{3}{4}=\ln\left(\frac{3}{4}+\sqrt{\frac{9}{16}+1}\right)=\ldots$	**M1**	Uses $\mathrm{arsinh}\,t=\ln\left(t+\sqrt{t^2+1}\right)$ on at least one of their values, or evaluates a logarithmic form from (b) at a limit.
	$\frac{1}{6}(\ln 2+\ln 2)=\frac{1}{3}\ln 2$	**A1**	cso. Any form $A\ln B$ with B an integer, such as $\frac{1}{6}\ln 4$. Not $\frac{1}{6}\left(\ln 2-\ln\frac{1}{2}\right)$.
		[3]	
	Total for Question 7	**7**	

Question	Scheme	Marks	Notes
8(a)	$\int \ln(1+x)\,\mathrm{d}x = x\ln(1+x) - \int \frac{x}{1+x}\,\mathrm{d}x$	**M1**	Applies parts with $u = \ln(1+x)$, reaching $px\ln(1+x) - \int \frac{qx}{1+x}\,\mathrm{d}x$, or the equivalent after substituting $t = 1+x$. Condone sign slips.
	$= x\ln(1+x) - x + \ln(1+x)$	**A1**	oe, such as $(1+x)\ln(1+x) - x$ or $t\ln t - t$.
	$\frac{1}{e-1}\Big[(1+x)\ln(1+x) - x\Big]_0^{e-1}$ $= \frac{1}{e-1}\big((e\ln e - (e-1)) - 1\times\ln 1\big)$	**dM1**	Substitutes both limits, or 1 and e after substituting $t = 1+x$, into their antiderivative and divides by $e-1$.
	$= \frac{1}{e-1}$	**A1 AG**	Correct substituted expression, using $\ln e = 1$ and $\ln 1 = 0$, leading without error to the printed result.
		[4]	
8(b)	$x = e^y - 1$	**B1**	
	$\frac{1}{1-0}\int_0^1 (e^y - 1)\,\mathrm{d}y = \Big[e^y - y\Big]_0^1 = \ldots$	**M1**	Integrates their x in terms of y, at least one term correct, and substitutes $y = 1$ and $y = 0$. Division by $1-0$ need not be shown.
	$e - 2$	**A1**	
		[3]	
	Total for Question 8	**7**	

Question	Scheme	Marks	Notes
9	$\frac{n}{2\pi}\int_0^{2\pi/n}\left(a^2\sin^2 nt + 2ab\sin nt\sin 2nt + b^2\sin^2 2nt\right)\mathrm{d}t$	**M1**	Forms the integral of I^2, expanded into three terms, from 0 to $\frac{2\pi}{n}$, with or without $\frac{n}{2\pi}$. Condone a slip in the cross term's coefficient.
	$\int\sin^2 nt\,\mathrm{d}t = \int\frac{1}{2}(1-\cos 2nt)\,\mathrm{d}t = \frac{t}{2} - \frac{\sin 2nt}{4n}$	**M1**	Uses $\sin^2 x = \frac{1}{2}(1\pm\cos 2x)$ for at least one square and integrates to the form $pt + q\sin knt$. Condone sign and coefficient slips.
	$\int_0^{2\pi/n}\sin^2 nt\,\mathrm{d}t = \int_0^{2\pi/n}\sin^2 2nt\,\mathrm{d}t = \frac{\pi}{n}$	**A1**	
	$\int\sin nt\sin 2nt\,\mathrm{d}t = \int 2\sin^2 nt\cos nt\,\mathrm{d}t = \frac{2\sin^3 nt}{3n}$	**M1**	Integrates the cross term by a valid method, e.g. to the form $k\sin^3 nt$, or by product-to-sum to $p\sin nt + q\sin 3nt$. Condone coefficient slips.
	$\left[\frac{2\sin^3 nt}{3n}\right]_0^{2\pi/n} = 0$	**A1**	Limits substituted into a correct integral; a bare statement that the integral is zero does not score.
	$\frac{n}{2\pi}\left(a^2\times\text{“}\frac{\pi}{n}\text{”} + 2ab\times\text{“0”} + b^2\times\text{“}\frac{\pi}{n}\text{”}\right)$ $\Rightarrow \text{RMS} = \sqrt{\ldots}$	**dM1**	Combines their three integrals with their coefficients, multiplies by $\frac{n}{2\pi}$ and takes the square root. Depends on the first M1.
	$\frac{a^2+b^2}{2} \Rightarrow \text{RMS} = \sqrt{\frac{a^2+b^2}{2}}$	**A1 AG**	cso: correct integrals combined without error to the printed result.
	Total for Question 9	**7**	

Question	Scheme	Marks	Notes
10	$\dfrac{x+3}{x^2(x+1)} = \dfrac{A}{x} + \dfrac{B}{x^2} + \dfrac{C}{x+1}$	**M1**	Uses partial fractions with separate terms in x and x^2 for the repeated factor, and a term in $x+1$.
	$x+3 = Ax(x+1) + B(x+1) + Cx^2$ $\Rightarrow A = \ldots,\ B = \ldots,\ C = \ldots$	**dM1**	Clears fractions and finds all three constants, by substitution or equating coefficients. Condone arithmetic slips.
	$-\dfrac{2}{x} + \dfrac{3}{x^2} + \dfrac{2}{x+1}$	**A1**	
	$\left[-2\ln x - \dfrac{3}{x} + 2\ln(x+1)\right]$	**M1 A1**	**M1** Integrates their partial fractions to the form $p\ln x + \frac{q}{x} + r\ln(x+1)$. Condone sign and coefficient slips. **A1** Correct antiderivative; modulus signs optional.
	$\lim_{t\to\infty}\left[2\ln\left(\dfrac{x+1}{x}\right) - \dfrac{3}{x}\right]_1^t$ $= \lim_{t\to\infty}\left(2\ln\left(1+\frac{1}{t}\right) - \frac{3}{t} - 2\ln 2 + 3\right)$	**M1**	Uses an upper limit t, combines their two log terms into one logarithm, before or after substituting, and substitutes both limits. Not if ∞ is substituted.
	$t\to\infty:\ \ln\left(1+\frac{1}{t}\right)\to\ln 1 = 0,\ \frac{3}{t}\to 0$	**dM1**	States, as $t\to\infty$, that their combined log term tends to $\ln 1 = 0$ and their $\frac{k}{t}$ term tends to 0.
	$\displaystyle\int_1^\infty \frac{x+3}{x^2(x+1)}\,\mathrm{d}x = 3 - 2\ln 2 = 3 - \ln 4$	**A1 AG**	cso, from correct limits leading without error to $3-\ln 4$. Withhold if ∞ is substituted or $\ln\infty$ terms are cancelled anywhere.
	Total for Question 10	**8**	

Question	Scheme	Marks	Notes
11(a)	Upper limit is infinite, so the interval of integration is unbounded	**E1**	States that the upper limit is infinite, or the interval is unbounded. Not for a reference to the integrand alone.
		[1]	
11(b)	$\frac{10}{(x+2)(3x+1)} = \frac{A}{x+2} + \frac{B}{3x+1}$ $10 = A(3x+1) + B(x+2) \Rightarrow A = \ldots,\ B = \ldots$	**M1**	Uses the correct partial-fraction form and a method for both constants, by substitution or comparing coefficients.
	$A = -2,\ B = 6$	**A1**	
	$\int \left(-\frac{2}{x+2} + \frac{6}{3x+1}\right) \mathrm{d}x$ $= p\ln(x+2) + q\ln(3x+1)$	**M1**	Integrates their two partial fractions to the form $p\ln(x+2) + q\ln(3x+1)$. Condone coefficient slips and missing modulus signs.
	$= 2\ln\left(\frac{3x+1}{x+2}\right)$	**A1**	Correct single logarithm, oe; may first be seen after the limits are substituted.
	$\left[2\ln\frac{3x+1}{x+2}\right]_0^t = 2\ln\frac{3t+1}{t+2} - 2\ln\frac{1}{2}$ $\frac{3t+1}{t+2} = \frac{3+\frac{1}{t}}{1+\frac{2}{t}} \to \ldots$ as $t \to \infty$	**dM1**	Replaces ∞ by a variable, substitutes both limits and finds the limit of their logarithm's argument as it tends to infinity. Not for substituting ∞ directly.
	$2\ln 3 + 2\ln 2 = 2\ln 6$	**A1**	cso. Any correct $a\ln b$ with integer a and rational b, or $a = 2$, $b = 6$ stated.
		[6]	
	Total for Question 11	**7**	

Question	Scheme	Marks	Notes
12(a)	At $x = 1$, $\sqrt[3]{x^2 - 1} = 0$: integrand undefined $x = 1$ lies in $[0, 3]$	**E1**	States the integrand is undefined or unbounded (or denominator zero) at $x = 1$, and that 1 lies within the interval of integration.
		[1]	
12(b)	$\int x\left(x^2 - 1\right)^{-\frac{1}{3}}\,\mathrm{d}x = k\left(x^2 - 1\right)^{\frac{2}{3}}$	**M1**	Integrates to the form $k\left(x^2 - 1\right)^{\frac{2}{3}}$, or $ku^{\frac{2}{3}}$ with $u = x^2 - 1$, by inspection or substitution.
	$= \frac{3}{4}\left(x^2 - 1\right)^{\frac{2}{3}}\ (+c)$	**A1**	oe, including $\frac{3}{4}u^{\frac{2}{3}}$ with $u = x^2 - 1$.
	Splits at $x = 1$: $\lim_{t\to 1^-}\left[\frac{3}{4}\left(x^2 - 1\right)^{\frac{2}{3}}\right]_0^t$ $+ \lim_{s\to 1^+}\left[\frac{3}{4}\left(x^2 - 1\right)^{\frac{2}{3}}\right]_s^3$	**M1**	Splits at $x = 1$ into two integrals, each written with a limit of a variable endpoint tending to 1 from the appropriate side.
	$= \left(0 - \frac{3}{4}\right) + (3 - 0)$	**dM1**	Substitutes the endpoints into their antiderivative in both integrals and evaluates the limits, with the terms at the variable endpoint tending to 0.
	Both limits finite, so converges; $= \frac{9}{4}$	**A1**	cso. $\frac{9}{4}$ oe with a statement that the integral converges or both limits are finite. A0 without the limit argument.
		[5]	
	Total for Question 12	**6**	

Question	Scheme	Marks	Notes
13(a)	Integrand undefined (unbounded) at the lower limit $x = 0$	**E1**	States the integrand is undefined, or tends to infinity, at $x = 0$. Not for a reason based on the range or the upper limit.
		[1]	
13(b)	$\int x^{-\frac{1}{2}}(\ln x)^2\,\mathrm{d}x = 2x^{\frac{1}{2}}(\ln x)^2$ $- \int 2x^{\frac{1}{2}} \times \frac{2\ln x}{x}\,\mathrm{d}x$	**M1 A1**	**M1** Integrates by parts with $u = (\ln x)^2$, reaching the form $px^{\frac{1}{2}}(\ln x)^2 - q\int x^{-\frac{1}{2}}\ln x\,\mathrm{d}x$. Condone coefficient slips. **A1** Correct; need not be simplified.
	$\int x^{-\frac{1}{2}}\ln x\,\mathrm{d}x = 2x^{\frac{1}{2}}\ln x - \int 2x^{-\frac{1}{2}}\,\mathrm{d}x$	**dM1**	Integrates $\int x^{-\frac{1}{2}}\ln x\,\mathrm{d}x$ by parts with $u = \ln x$, reaching the form $rx^{\frac{1}{2}}\ln x - s\int x^{-\frac{1}{2}}\,\mathrm{d}x$. Condone coefficient slips.
	$= 2\sqrt{x}(\ln x)^2 - 4\left(2\sqrt{x}\ln x - 4\sqrt{x}\right) + c$ $= 2\sqrt{x}\left((\ln x)^2 - 4\ln x + 8\right) + c$	**A1 AG**	Both stages correctly combined, final integral done, leading without error to the printed result, including $+c$.
		[4]	
13(c)	$\lim_{t\to 0^+}\int_t^e x^{-\frac{1}{2}}(\ln x)^2\,\mathrm{d}x$ $= \lim_{t\to 0^+}\left[2\sqrt{x}\left((\ln x)^2 - 4\ln x + 8\right)\right]_t^e$	**M1**	Replaces the lower limit by t, or another letter, and substitutes both limits into the result of (b). Not for substituting $x = 0$.
	$= 10\sqrt{e} - 2\sqrt{t}\left((\ln t)^2 - 4\ln t + 8\right)$	**A1**	Correct expression in t, oe, e.g. expanded.
	As $t \to 0^+$: $\sqrt{t}(\ln t)^2 \to 0$, $\sqrt{t}\ln t \to 0$, $\sqrt{t} \to 0$	**dM1**	Lets $t \to 0^+$, using the given result for the $\sqrt{t}(\ln t)^2$ and $\sqrt{t}\ln t$ terms and $\sqrt{t} \to 0$.
	Limit finite, so integral exists, $= 10\sqrt{e}$	**A1**	cso. Exact $10\sqrt{e}$ with a statement that the limit is finite, or the integral exists.
		[4]	
	Total for Question 13	**9**	

Question	Scheme	Marks	Notes
14(a)	$\int \frac{(\ln x)^2}{x^2}\,\mathrm{d}x = -\frac{(\ln x)^2}{x} + \int \frac{2\ln x}{x^2}\,\mathrm{d}x$	**M1 A1**	**M1** Integrates by parts with $u = (\ln x)^2$ (or t^2 after $x = e^t$) to the form $p\frac{(\ln x)^2}{x} + q\int \frac{\ln x}{x^2}\,\mathrm{d}x$. Condone sign and coefficient slips. **A1** Correct, need not be simplified.
	$\int \frac{\ln x}{x^2}\,\mathrm{d}x = -\frac{\ln x}{x} + \int \frac{1}{x^2}\,\mathrm{d}x$	**dM1**	Integrates $\frac{\ln x}{x^2}$ (or te^{-t}) by parts to the form $p\frac{\ln x}{x} + q\int \frac{1}{x^2}\,\mathrm{d}x$. Condone sign and coefficient slips.
	$-\frac{(\ln x)^2}{x} - \frac{2\ln x}{x} - \frac{2}{x}$	**A1**	oe; constant not needed.
	$\left(-\frac{(\ln a)^2}{a} - \frac{2\ln a}{a} - \frac{2}{a}\right) - (-2)$	**dM1**	Substitutes a and 1 into their antiderivative from two integrations by parts and subtracts.
	$2 - \frac{(\ln a)^2}{a} - \frac{2\ln a}{a} - \frac{2}{a}$	**A1**	oe
		[6]	
14(b)	0	**B1**	
		[1]	
14(c)	$\lim_{a\to\infty}\left(2 - \frac{(\ln a)^2}{a} - \frac{2\ln a}{a} - \frac{2}{a}\right)$ $= 2 - 0 - 0 - 0$	**M1**	Lets $a \to \infty$ in their (a) of the form constant plus multiples of $\frac{(\ln a)^2}{a}$, $\frac{\ln a}{a}$, $\frac{1}{a}$, stating each such term tends to 0.
	$\int_1^\infty \frac{(\ln x)^2}{x^2}\,\mathrm{d}x = 2$	**A1**	
		[2]	
Total for Question 14		**9**	

Question	Scheme	Marks	Notes
15	$\dfrac{5x-3}{(x^2+3)(x+1)} = \dfrac{A}{x+1} + \dfrac{Bx+C}{x^2+3}$	**M1**	Uses a constant over $x+1$ and a linear numerator over x^2+3.
	$5x-3 = A(x^2+3)+(Bx+C)(x+1)$ $\Rightarrow A = \ldots,\ B = \ldots,\ C = \ldots$	**dM1**	Multiplies through and finds all three constants by substitution or equating coefficients. Condone arithmetic slips.
	$A=-2,\ B=2,\ C=3$	**A1**	
	$\int \left(-\dfrac{2}{x+1} + \dfrac{2x}{x^2+3}\right) \mathrm{d}x$ $= -2\ln(x+1) + \ln(x^2+3)$	**M1**	Integrates their $\frac{A}{x+1}$ and $\frac{Bx}{x^2+3}$ to the forms $\alpha\ln(x+1)$ and $\beta\ln(x^2+3)$. Condone coefficient slips.
	$\int \dfrac{3}{x^2+3}\,\mathrm{d}x = \dfrac{3}{\sqrt{3}}\arctan\left(\dfrac{x}{\sqrt{3}}\right)$	**M1**	Integrates their $\frac{C}{x^2+3}$ to the form $k\arctan\left(\frac{x}{\sqrt{3}}\right)$.
	$\ln\left(\dfrac{x^2+3}{(x+1)^2}\right) + \sqrt{3}\arctan\left(\dfrac{x}{\sqrt{3}}\right)$	**A1**	Correct antiderivative oe, logarithms combined or not; modulus signs condoned.
	$\lim_{t\to\infty}\left[\ln\left(\dfrac{t^2+3}{(t+1)^2}\right) + \sqrt{3}\arctan\left(\dfrac{t}{\sqrt{3}}\right)\right]$ $-(\ln 3 + 0)$ $= 0 + \sqrt{3}\cdot\dfrac{\pi}{2} - \ln 3$	**dM1**	Replaces ∞ by a limit, applies both limits, combines the logarithms into one tending to $\ln 1$, and uses $\arctan \to \frac{\pi}{2}$. Not for evaluating separate logarithms at infinity.
	$\frac{1}{2}\sqrt{3}\,\pi + \ln\frac{1}{3}$: $p=\frac{1}{2},\ q=\frac{1}{3}$	**A1**	cso. Needs $q=\frac{1}{3}$ or $\ln\frac{1}{3}$; $-\ln 3$ alone scores A0.
	Total for Question 15	**8**	

Question	Scheme	Marks	Notes
16(a)	As $y \to 0^+$, $\ln y \to -\infty$, $(\ln y)^2 \to \infty$ Integrand unbounded at $y = 0$	**E1**	States the integrand is unbounded or undefined at $y = 0$, referring to $\ln y$ or $(\ln y)^2$ there.
		[1]	
16(b)(i)	$(\ln y)^2 = x^2$, $\mathrm{d}y = -e^{-x}\,\mathrm{d}x$	**M1**	Replaces $(\ln y)^2$ by x^2 and $\mathrm{d}y$ by $\pm e^{-x}\,\mathrm{d}x$. Condone a sign slip in $\mathrm{d}y$.
	$y = 1 \Rightarrow x = 0$ $y \to 0^+ \Rightarrow x = -\ln y \to \infty$	**B1**	Both limits, with $x \to \infty$ justified as $y \to 0^+$ by $-\ln y \to \infty$ or $e^{-x} \to 0$. B0 for $y = 0 \Rightarrow x = \infty$ with no limiting argument.
	$\int_{\infty}^{0} x^2\,(-e^{-x})\,\mathrm{d}x = \int_{0}^{\infty} x^2 e^{-x}\,\mathrm{d}x$	**A1 AG**	Correct substituted integral with limits ∞ to 0 and the minus sign, reversed to the printed result without error.
		[3]	
16(b)(ii)	$\int x^2 e^{-x}\,\mathrm{d}x = -x^2 e^{-x} + \int 2x e^{-x}\,\mathrm{d}x$	**M1 A1**	**M1** Integrates by parts with $u = x^2$, reaching $\pm x^2 e^{-x} \pm \int kx e^{-x}\,\mathrm{d}x$. Tabular method accepted.
	$= -x^2 e^{-x} - 2x e^{-x} - 2e^{-x}$	**dM1 A1**	**dM1** Integrates their $\int kx e^{-x}\,\mathrm{d}x$ by parts, reaching the form $px e^{-x} + qe^{-x}$. **A1** oe; constant not needed.
	$\left[-e^{-x}(x^2 + 2x + 2)\right]_0^t = 2 - e^{-t}(t^2 + 2t + 2)$ As $t \to \infty$, $t^2 e^{-t}, te^{-t}, e^{-t} \to 0$	**dM1**	Applies limits 0 and t to their antiderivative of the form $e^{-x} \times$ quadratic, and lets $t \to \infty$ with the exponential terms tending to 0. Not for substituting ∞ directly.
	$\int_{0}^{\infty} x^2 e^{-x}\,\mathrm{d}x = 2$	**A1**	cso. A bare answer scores nothing.
		[6]	
16(b)(iii)	$\int_{0}^{1} (\ln y)^2\,\mathrm{d}y = 2$	**B1ft**	ft their value from (b)(ii).
		[1]	
	Total for Question 16	**11**	